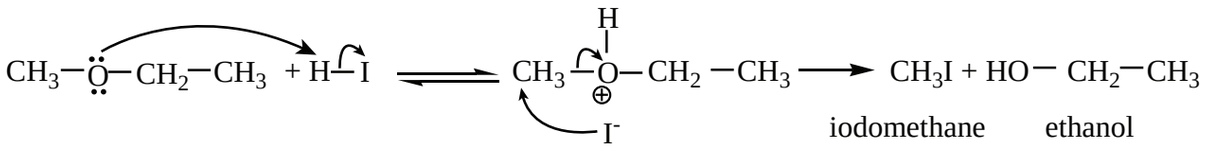
Mechanism:

Ethers having primary alkyl group undergo SN^2 reaction while tertiary alkyl ether undergo SN^1 reaction.Protonation of ether is followed by the attack of halide ion.The halide ion preferentially attacks the less sterically hindered of the two alkyl groups which are attached to etherial oxygen.



When excess HBr or HI is used, the alcohol formed will further react with HBr or HI to form alkyl halides.

$$CH_3\text{-}CH_2\text{-}OH \xrightarrow{\text{excessHBr}} CH_3\text{-}CH_2\text{-}Br$$

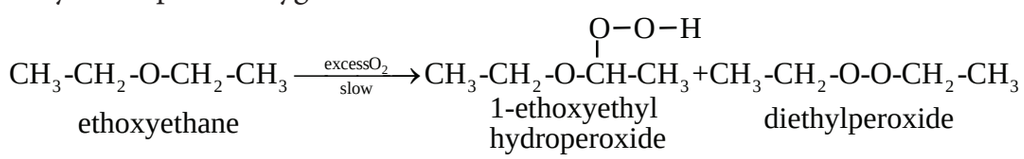
ethanol — bromoethane

Evaluate Yourself:

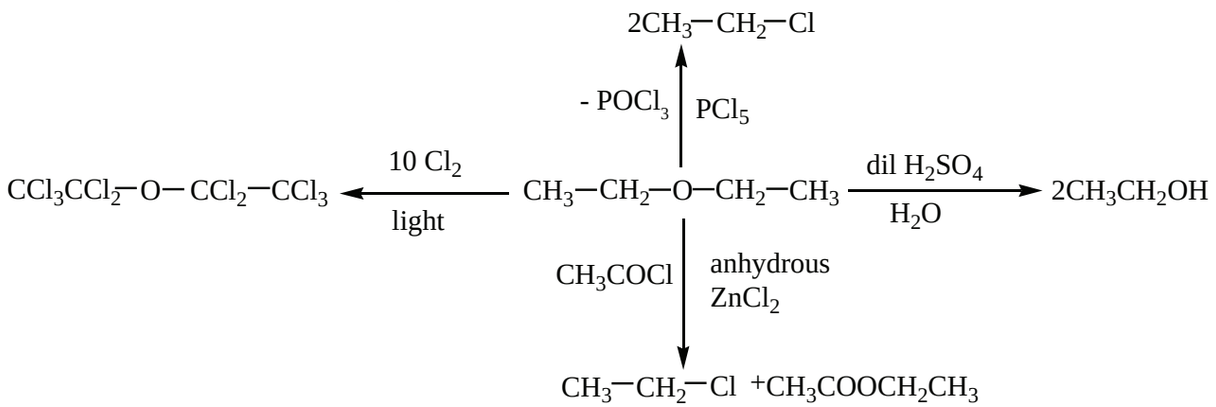
1mole of HI is allowed to react with t-butyl methylether. Identify the product and write down the mechanism of the reaction.

Autooxidation of ethers:

When ethers are stored in the presence of atmospheric oxygen, they slowly oxidise to form hydroperoxides and dialkylperoxides. These are explosive in nature. Such a spontaneous oxidation by atmospheric oxygen is called autooxidation.



Some of the reaction of diethyl ether.



Aromatic electrophilic substitution reactions:

The alkoxy group (-OR) is an ortho, para directing group as well as activating group.It activates the aromatic ring towards electrophilic substitution.

XII U11-Hydroxy compounds.indd 137 11-01-2021 15:45:06